Motor-imagery EEG signal decoding using multichannel-empirical wavelet transform for brain computer interfaces

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***Abstract*—Motor-imagery (MI) electroencephalography (EEG) signal decomposition is an emerging technique for improving the performance of brain computer interfaces (BCIs). We proposed a multichannel-empirical wavelet transform (EWT) representation combined with a scattering convolution network (SCN) to efficiently decode the brain activity and extract relevant wave patterns for MI-based BCI. Two different preprocessing steps were tested: the first (PM1) included a bandpass Butterworth filter (1-40 Hz) and the independent component analysis (ICA), the second one (PM2) consisted only of a bandpass Butterworth filter (8-30 Hz). A binary support vector machine (SVM) classifier was used and the performance was evaluated in terms of classification accuracy. The proposed framework was assessed using the BCI competition IV dataset IIa, which contains EEG from 9 healthy subjects. PM1 presented a maximum mean accuracy over all subjects of 82.05% in the classification of the tongue and the left-hand MI tasks. PM2 achieved an average accuracy over all subjects of 88.40% and a standard deviation of 3.01 outperforming other state of the art methods in classifying right-hand and left-hand MI tasks. Finally, we observed that the best channels, intended as the channels holding the highest discrimination power between two MI tasks, were highly subject-specific and thus enabling task-based channel selection is crucial.**

***Index Terms*—Motor-imagery brain computer interface (MI-BCI), empirical wavelet transform (EWT), scattering convolution network (SCN), support vector machine (SVM).**

I. Introduction

A brain computer interface (BCI) is a system that is capable of extracting human intents from changes in brain activity and translating them into a control signal in real time [1]. During the last decades, BCI devices found applications in clinical and biomedical fields proving to be useful for diagnosis, monitoring, prevention and rehabilitation. In fact, a BCI system can help disabled people to interact with the external world only via thoughts, to control prosthetic limbs or to recover from stroke and restore lost physical functions. Since many other applications out of the clinical field are emerging (e.g., smart environment, games and entertainment, etc.) even healthy people can take advantage of the system [2], [3].

Electroencephalography (EEG) is the most used signal acquisition technique in the field of BCI [4] and measures the brain activity non-invasively from the scalp [5]. The EEG signal has an excellent time resolution; however, it is non-stationary, highly subject-specific and noisy. For instance, many physiological artifacts (e.g., lateral-eye movement, eye blinks, cardiac and muscle activity) can occur in EEG signal and interfere with neural activity. A typical non-invasive and endogenous EEG BCI system is based on the motor imagery (MI), which can be defined as the mental imagination of a motor act without the actual execution of the movement [1]. Those MI-BCI systems monitor the sensorimotor rhythms (SMRs), which are oscillatory events in the EEG signals, from the primary sensorimotor (SMI) cortex. In particular, the movement, its preparation or the imagination of the same movement is accompanied by a decrease in μ rhythm. This phenomenon is called event-related desynchronization (ERD).

A general BCI signal processing pipeline consists of three main steps: 1) preprocessing, 2) feature extraction and 3) classification [1]. Data *preprocessing* is a critical step which could affect the entire pipeline. In general, temporal filtering (e.g. bandpass, Notch or Laplace) and/or spatial filtering (e.g. common average reference [CAR] and independent component analysis [ICA]) are commonly used to reduce the EEG noise and artifacts [1]. One of the most important components of a BCI system is *feature extraction*. Many methods have been implemented for extracting only the most important information. Among them, the Fourier transform, autoregressive models and common spatial patterns (CSP) are very popular, but are not able to provide any information on the temporal evolution of the EEG signals or can suffer from the classification success rates [6]. More recently, many signal decomposition (SD) methods, that generate several sub-band signals, have been used. The wavelet transform, the empirical mode decomposition [7], the wavelet packet decomposition and the empirical wavelet transform (EWT) are some good examples [8]. Sadiq

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et al. [9] provided evidence of the great potential of EWT, an adaptive SD method, for MI-EEG task classification. EWT produces many features well-localized in time and frequency domains. However, they are not translation-invariant (e.g. if translation is applied, the wavelet coefficients will change). To solve this issue, Priya et al. [10], using a cognitive task, combined EWT with a scattering convolution network (SCN), ensuring translation invariance of the extracted features up to a predefined scale. SCNs hold the same structure as convolutional neural networks (CNN) with the advantages of having a reduced computational complexity, employing more interpretable features and requiring few parameters. Regarding the *classification* step, traditional machine learning techniques, which demonstrated good performance, are still widely used (e.g., support vector machine [SVM], linear discriminant analysis and K-nearest neighbor) [4].

This work aims at exploring the capability of multichannel-EWT combined with the SCN in decoding the brain wave pattern modulated during a MI task using the SVM classifier.

II. METHODS

A. Dataset and acquisition protocol

In order to evaluate the proposed method, we used the publicly available EEG dataset from the BCI competition IV dataset IIa [11]. It contains data recording from nine controls while performing four MI tasks (i.e., left-hand [L], right-hand [R], both feet [F] and tongue [T]). For each subject, two sessions on different days were recorded, for a total of 288 trials per session (72 cue-based trials for each of the four classes). EEG data were recorded using a 22-channel EEG system at a sampling frequency of 250 Hz. Channel locations are reported in Fig. 1. In our case, the time interval was selected between 1 s and 3 s after the cue-onset, from which the subject has imagined the movement.

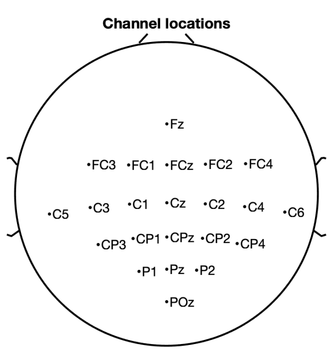

Fig. 1. Montage of the 22-channel EEG system.

B. Preprocessing

Data preprocessing is a critical step in BCI to achieve good classification performance. As discussed above, EEG signals are corrupted by artifacts. To overcome this issue, two preprocessing methods were considered: zero-phase Butterworth filtering between 1 and 40 Hz followed by ICA, referred to as PM1, and zero-phase Butterworth filtering between 8 and 30 Hz (PM2). The filtering was repeated twice, in forward and reverse directions on the signal time history, respectively, to obtain a zero-phase filter. To reduce the effect of temporal drifts, a baseline correction was performed and the EEG signals were re-referenced to CAR [1]. The EEG was filtered between 1 and 40 Hz (PM1) or between 8-30 Hz (PM2), in order to preserve both α (8-12 Hz) and β (13-30 Hz) frequency bands which are relevant for MI tasks, although in the first case (PM1) some physiological artifacts arising from sources other than the brain were still visible. To reduce the unwanted noise in PM1, ICA was applied, which has been extensively used for preprocessing and enhancing the relevant information embedded in the EEG signals [12].

C. Feature extraction

EWT, an extension of classical wavelet, is an adaptive method which aims at decomposing a non-stationary signal into different components, usually called "modes". This self-adapting method builds a wavelet filter bank, that is composed by a set of bandpass filters, whose support in the Fourier domain depends on where the information is located in the spectrum. For more details we refer to [13].

The wavelet transform is well-known for its ability of feature localization in time and frequency domains simultaneously, but it is not translation-invariant weakening the discriminative potential of the extracted features. To solve this issue wavelet SCN can be used.

A scattering transform computes higher order coefficients by iterating on wavelet transforms and modulus operators. Wavelet coefficients are computed up to a maximum scale 2^J (i.e. called invariance scale) and the lower frequencies are filtered by $\Phi_{2^J}(u)$ [14]. A SCN shares the structure of the traditional deep CNN, where convolution filters correspond to wavelet filters, the rectified linear unit (ReLU) to the modulus operator and the pooling relies on the low-pass filtering $\Phi_{2^J}(u)$. The benefits of SCN are: filters are not learned from data, but are predefined wavelets, such that the computational complexity is reduced and the extracted features are invariant up to a given scale. Moreover, the scattering transform preserves the signal energy [14].

The invariance scale was fixed to 2s (i.e. the length of the signal), which improved the subsequent classification performance and, thus, we set the SCN hyperparameter to that value. The number of modes was limited to 6 to ensure a similar feature extraction framework across epochs. The reason for choosing 6 modes was that this number allows to separate the frequency range in the main frequency bands. The translation invariant features (TIF) were then derived separately for each epoch, mode and channel.

All EEG channels were considered instead of choosing only one or few of them, because this helped to capture the interrelationships between the brain regions. Noteworthy, in this work the channel to be retained for classification was not fixed *a-priori*, as it is usual in the state of the art (SOA). Instead, all the channels were put in competition and the one leading to the best classification performance was retained.

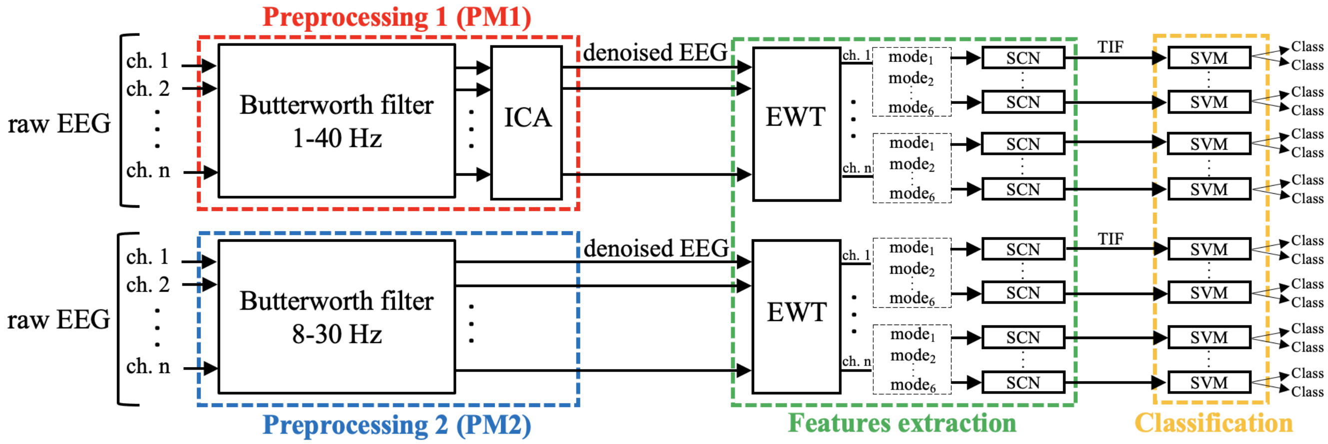

Fig. 2. Block diagram of the proposed framework.

D. Classification

In this work, we used a SVM since it is one of the most common classification algorithms for MI-EEG based BCI and its performance is superior than other methods [4]. The radial basis function (RBF) kernel was selected and the hyperparameters (i.e. C and γ) tuning was done by grid search. The mean accuracy values over a 5-fold cross-validation (CV) was calculated. We focused only on binary classification of six pairs of MI tasks: R vs. L, T vs. L, T vs. R, T vs. F, F vs. L, F vs. R. According to the classifier of interest, the TIF derived from the modes involved (as detailed in Section II-C) were used as input for the SVM classifier. For each classifier, the mode and the EEG channel leading to the best classification performance in the 5-fold CV were subject-specific.

Fig. 2 shows the block diagram of the proposed framework for subject-specific MI-EEG based BCI. We performed steps in Sections II-B and II-C in MATLAB R2021b and steps in Section II-D in Python.

TABLE I
VALIDATION ACCURACY OF PM1 FOR ALL BINARY CLASSIFICATIONS

	Accuracy (in %)					
Subj.	**T vs L**	**T vs R**	**T vs F**	**F vs R**	**F vs L**	**R vs L**
1	90.42	88.99	72.59	89.07	86.83	68.15
2	67.41	68.68	80.66	84.44	79.86	66.98
3	86.72	84.44	75.16	74.59	71.85	71.38
4	74.25	67.30	69.15	67.09	70.52	70.49
5	80.00	70.68	64.70	71.37	76.77	68.98
6	74.31	71.86	71.65	71.82	75.24	71.74
7	86.22	88.36	71.61	83.48	74.55	70.74
8	84.36	76.87	74.99	68.46	71.54	75.70
9	94.74	79.79	78.53	70.87	82.49	81.09
Mean	82.05	77.43	73.23	75.69	76.63	71.69

III. RESULTS

In Table I the validation accuracy values, obtained by classifying all the pairs of MI tasks following PM1, are provided. As shown, the results strongly depended on the subject and the two MI-tasks considered. The highest mean accuracy across subjects was 82.05% in T vs. L classification while the lowest was 71.69% in R vs. L classification.

TABLE II
COMPARISON BETWEEN PM1, PM2 AND EXISTING METHODS R VS L CLASSIFICATION PERFORMANCES

	Accuracy (in %)				
Subj.	**PM1**	**PM2**	**Tiwari et al. [7]**	**Zhang et al. [15]**	**Singh et al. [16]**
1	68.15	88.87	89.79	87.00	91.61
2	66.98	86.26	94.18	64.70	63.28
3	71.38	86.14	78.92	93.80	97.20
4	70.49	91.00	94.01	74.30	72.91
5	68.98	84.76	71.32	90.40	64.08
6	71.74	86.14	86.71	63.90	69.71
7	70.74	89.48	89.36	91.40	81.25
8	75.70	94.44	82.11	95.60	96.52
9	81.09	88.56	86.18	81.30	92.30
Mean	71.69	**88.40**	85.84	82.49	80.98
SD	4.31	**3.01**	7.40	12.21	13.86

TABLE III
BEST CHANNELS RESULTED AFTER PM2 FOR ALL BINARY CLASSIFICATIONS AND FOR EACH SUBJECT

Subj.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
R vs L	P1	C5	P1	CP4	**POz**	Pz	CP1	FCz	CP2
T vs L	**POz**	C5	Pz	**POz**	CPz	**Cz**	CP2	CP3	C4
T vs R	P1	**Cz**	FC2	**Cz**	P2	C2	FCz	CPz	C2
T vs F	Pz	CP1	CP3	**Cz**	P1	**Cz**	CP2	CP4	FCz
F vs R	CP3	CP1	**POz**	**Cz**	C4	FC4	FC4	C6	P1
F vs L	CP2	**POz**	C6	C1	**POz**	C1	CP4	Pz	**POz**

In Table II, the accuracy values reached after PM1 and PM2 were compared with other SOA methods [7], [15], [16] only for R vs. L classification. The PM2 provided an improvement in the performance compared to PM1 and existing SOA results (with an average accuracy of 88.40% and a standard deviation [SD] of 3.01). Of note, the accuracy reached for subjects 2 and 6 obtained after PM1 was higher than that reported in Zhang et al. [15] and Singh et al. [16] works, while the accuracy for subject 4 was comparable with those studies. Moreover, the standard deviations of the proposed methods are lower than the SOA methods. In [16] there are other SOA results for R vs. L classification and the average accuracy of the PM2 is always higher.

For each subject and binary classifications, table III shows the channel that leads to best accuracy using PM2. We can observe that for the same task combination the results can be very variable. However, the most frequently channel, identified as the best ones, were Cz and POz, which are located in the central and the post-central areas, respectively.

IV. Discussion and Conclusion

A multichannel-EWT combined with SCN provided a clear improvement in the performance of the MI-BCI pipeline. In the proposed framework (EWT+SCN) only few parameters need to be controlled (i.e., the numbers of modes and the invariance scale) while the signal decomposition is completely automatic. Before analyzing the data with the proposed framework, we tested two commonly used preprocessing pipelines (PM1 and PM2) to improve the initial filtering step as well. At the same time, the filtering enhanced the informative features of the EEG signal with respect to the others, improving its interpretability.

PM2 outperformed existing SOA results with a mean validation accuracy across subjects of 88.40% and a standard deviation of 3.01. For instance, Tiwari et al. [7] used a multivariate-empirical mode decomposition to extract features, which are much more complex to interpret than EWT. Zhang et al. [15] defined a new algorithm for the simultaneous optimization of filter bands and time window within CSP, but its parameter selection was time-consuming and the approach required another dataset for validation. Finally, Singh et al. [16] used both Euclidean and Riemannian approaches, but also in this case the analysis was based on spatial filters.

We have previously discussed how the preprocessing step can be crucial and how many choices (e.g. the frequency range of the bandpass filter, and the ICA that is highly operator-dependent) could have a strong impact on classification results. Despite this, the peak variability of the power spectrum in a specific frequency band [17] is intrinsically taken into account thanks to the EWT, that is an adaptive signal processing method.

A large number of channels can increase the computational complexity of the analysis and may affect the performance of real time applications. Nonetheless, the results reported in Table III confirmed the hypothesis that BCI is highly subject-specific and thus enabling task-based channel selection is crucial. From prior neurophysiological knowledge, the imagination of a specific task is well localized (e.g., C3 and C4 are principally referred to the imagination of the hand movements [18] and Cz to the feet MI task), but, despite this, the information coming from other EEG electrodes are also involved to better distinguish two MI tasks. This pushes towards the investigation of brain connectivity as a means to exploit the inter-dependencies across the EEG channels analyzing the coupling information or casual influence between EEG signals [19], [20].

In conclusion, we demonstrated the capability of multichannel-EWT combined with SCN to decode the brain wave during a MI task for BCI application and our future work will concern the investigation of brain connectivity.

References

[1] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, "Brain–computer interfaces for communication and control," *Clinical neurophysiology*, vol. 113, no. 6, pp. 767–791, 2002.

[2] V. Chamola, A. Vineet, A. Nayyar, and E. Hossain, "Brain-computer interface-based humanoid control: A review," *Sensors*, vol. 20, no. 13, p. 3620, 2020.

[3] J. N. Mak and J. R. Wolpaw, "Clinical applications of brain-computer interfaces: current state and future prospects," *IEEE reviews in biomedical engineering*, vol. 2, pp. 187–199, 2009.

[4] N. Padfield, J. Zabalza, H. Zhao, V. Masero, and J. Ren, "Eeg-based brain-computer interfaces using motor-imagery: Techniques and challenges," *Sensors*, vol. 19, no. 6, p. 1423, 2019.

[5] R. Abiri, S. Borhani, E. W. Sellers, Y. Jiang, and X. Zhao, "A comprehensive review of eeg-based brain–computer interface paradigms," *Journal of neural engineering*, vol. 16, no. 1, p. 011001, 2019.

[6] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, and F. Yger, "A review of classification algorithms for eeg-based brain–computer interfaces: a 10 year update," *Journal of neural engineering*, vol. 15, no. 3, p. 031005, 2018.

[7] A. Tiwari and A. Chaturvedi, "A novel channel selection method for bci classification using dynamic channel relevance," *IEEE Access*, vol. 9, pp. 126 698–126 716, 2021.

[8] J. Kevric and A. Subasi, "Comparison of signal decomposition methods in classification of eeg signals for motor-imagery bci system," *Biomedical Signal Processing and Control*, vol. 31, pp. 398–406, 2017.

[9] M. T. Sadiq, X. Yu, Z. Yuan, F. Zeming, A. U. Rehman, I. Ullah, G. Li, and G. Xiao, "Motor imagery eeg signals decoding by multivariate empirical wavelet transform-based framework for robust brain–computer interfaces," *IEEE access*, vol. 7, pp. 171 431–171 451, 2019.

[10] L. P. B. *et al.*, "Scattering convolutional network based predictive model for cognitive activity of brain using empirical wavelet decomposition," *Biomedical Signal Processing and Control*, vol. 66, p. 102501, 2021.

[11] C. Brunner, R. Leeb, G. Müller-Putz, A. Schlögl, and G. Pfurtscheller, "Bci competition 2008–graz data set a," *Institute for Knowledge Discovery (Laboratory of Brain-Computer Interfaces), Graz University of Technology*, vol. 16, pp. 1–6, 2008.

[12] M. K. Ghumman, S. Singh, N. Singh, and B. Jindal, "Optimization of parameters for improving the performance of eeg-based bci system," *Journal of Reliable Intelligent Environments*, vol. 7, no. 2, pp. 145–156, 2021.

[13] J. Gilles, "Empirical wavelet transform," *IEEE transactions on signal processing*, vol. 61, no. 16, pp. 3999–4010, 2013.

[14] J. Bruna and S. Mallat, "Invariant scattering convolution networks," *IEEE transactions on pattern analysis and machine intelligence*, vol. 35, no. 8, pp. 1872–1886, 2013.

[15] Y. Zhang, C. S. Nam, G. Zhou, J. Jin, X. Wang, and A. Cichocki, "Temporally constrained sparse group spatial patterns for motor imagery bci," *IEEE transactions on cybernetics*, vol. 49, no. 9, pp. 3322–3332, 2018.

[16] A. Singh, S. Lal, and H. W. Guesgen, "Small sample motor imagery classification using regularized riemannian features," *IEEE Access*, vol. 7, pp. 46 858–46 869, 2019.

[17] A. Goljahani, P. Bisiacchi, and G. Sparacino, "An eeglab plugin to analyze individual eeg alpha rhythms using the "channel reactivity-based method"," *Computer methods and programs in biomedicine*, vol. 113, no. 3, pp. 853–861, 2014.

[18] E. Formaggio *et al.*, "Modulation of event-related desynchronization in robot-assisted hand performance: brain oscillatory changes in active, passive and imagined movements," *Journal of neuroengineering and rehabilitation*, vol. 10, no. 1, pp. 1–10, 2013.

[19] L. Brusini, F. Stival, F. Setti, E. Menegatti, G. Menegaz, and S. F. Storti, "A systematic review on motor-imagery brain-connectivity-based computer interfaces," *IEEE Transactions on Human-Machine Systems*, 2021.

[20] S. F. Storti, E. Formaggio, P. Manganotti, and G. Menegaz, "Brain network connectivity and topological analysis during voluntary arm movements," *Clinical EEG and Neuroscience*, vol. 47, no. 4, pp. 276–290, 2016.